

AMRITA

Longevity medicine, calibrated for India.

Working name (alternates: **Ayuh Bio**, **Chiranya**, **GeroBharat**) — trademark search pending.

A computational drug-discovery company that screens already-approved Western medicines against the protein machinery of human ageing — to find cheap, safe, repurposable longevity drugs, calibrated to South-Asian biology — and proves they work by measuring biological age in real people.

1. The one-paragraph thesis

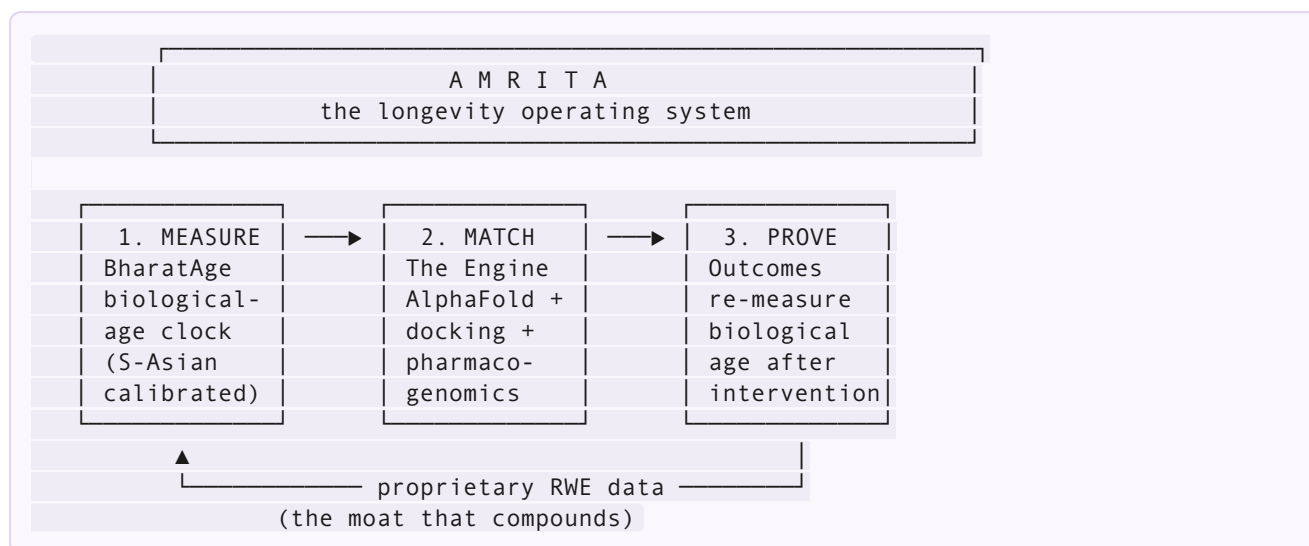
Ageing is now a *targetable* biological process, and the safest path to a longevity medicine is **repurposing drugs that are already approved** (metformin, rapamycin, SGLT2-inhibitors, senolytics). The discovery step — does an approved drug bind an ageing-related protein? — has become a **software problem** solvable with AlphaFold + molecular docking, on a laptop, for almost nothing. **Nobody has pointed this engine at South-Asian biology**, where people age on a different curve (the "thin-fat" phenotype, diabetes and heart disease a decade early). AMRITA builds that engine, calibrates it to Indian genetics, and closes the loop by measuring whether the drugs it surfaces actually reverse biological age in Indian patients — creating a proprietary dataset no competitor can copy.

2. Why now / why India (the funding wedge)

- **Why now:** AlphaFold (2021) + AlphaFold3 (2024) made protein structures free; docking + variant-effect AI matured; the first human longevity trials (TAME/metformin, rapamycin) made drug-repurposing for ageing scientifically respectable.
 - **Why India:**
 - **Different ageing biology** — South Asians develop insulin resistance, diabetes and coronary disease 5–10 years earlier, at lower BMI ("thin-fat" phenotype). Western drug-response data does not transfer.
 - **The world's pharmacy** — India manufactures generics at global scale; a repurposed geroprotector can be produced and distributed cheaply.
 - **Massive, under-studied population** — no omic-grade ageing biobank exists for 1.4 billion people. That gap is a globally valuable asset.
 - **Cost arbitrage** — dry-lab discovery + cheap diagnostics + cheap trials.
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3. What we are *ultimately* making — the platform

A three-layer loop. Each layer is a product; together they form an unfakeable data moat.



Layer 1 — MEASURE: *BharatAge* biological-age clock

Tells a person how fast they are *actually* ageing, calibrated to Indian physiology.

- Free/scale tier: face photo + voice + grip + questionnaire.
- Cheap tier: retinal photo (retinal-age gap predicts mortality).
- Premium tier: at-home dried-blood-spot → DNA-methylation + metabolomic clock.
- **Output:** biological age vs passport age, vs the *Indian* curve — not a Western one.

Layer 2 — MATCH: *The Engine* (today's proof-of-concept)

The computational discovery core.

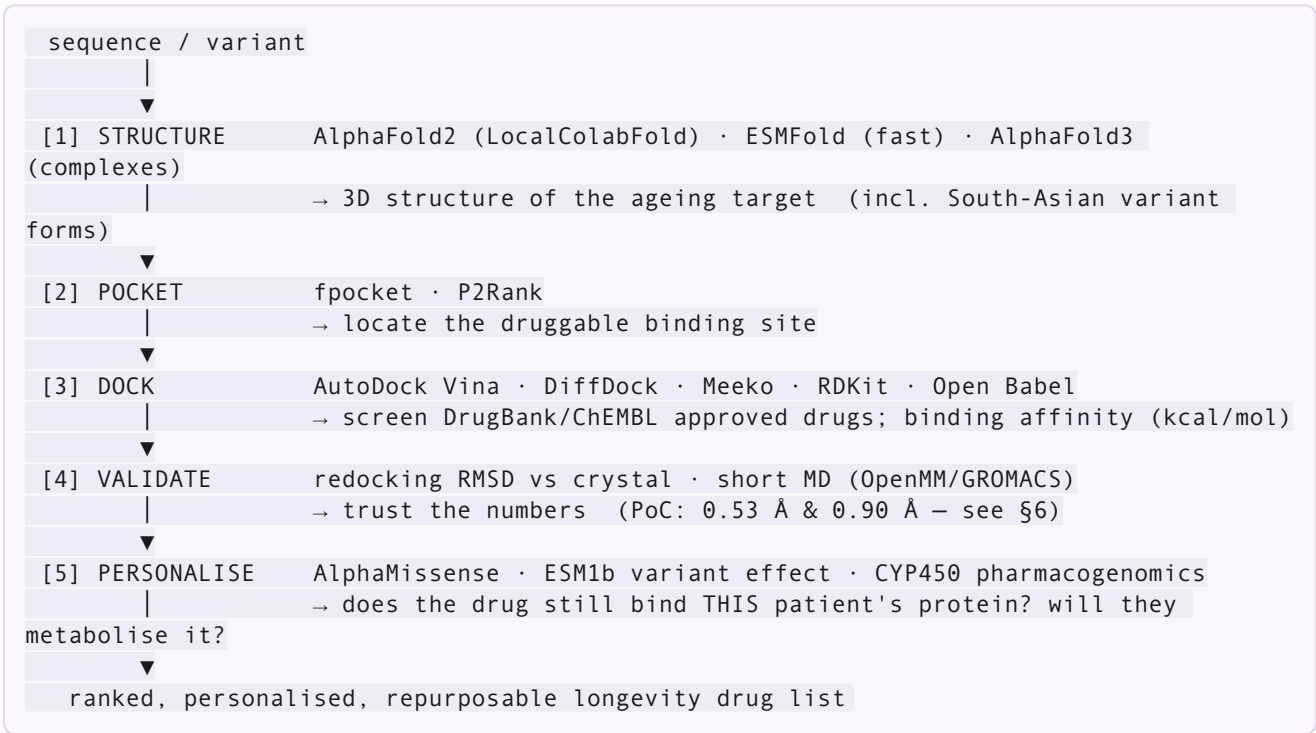
- Fold ageing-related target proteins (AlphaFold/ESMFold).
- Dock the library of **approved drugs** (DrugBank/ChEMBL) into their pockets.
- Rank repurposing hits; re-rank for the *individual's* genetic variants + drug-metabolism (CYP450) profile.
- **Output:** "for your ageing biology and genotype, these approved drugs best fit your geroprotective targets."

Layer 3 — PROVE: *Outcomes loop*

Physician-supervised, the user takes a matched approved drug; biological age is re-measured in 3–6 months.

- **Output:** real-world evidence — *which approved drug moved biological age, in which South-Asian phenotype.*
- **This dataset is the company.** It is proprietary, licensable to pharma, and gets better with every patient.

4. The science engine — architecture (the dry lab)



Ageing targets (starter set): mTOR, Bcl-2/Bcl-xL (senolysis), SIRT1–7, AMPK, NLRP3, KEAP1/NRF2, FOXO3, Klotho.

5. Tech stack (full)

Layer	Tools
Structure prediction	LocalColabFold (AlphaFold2), ESMFold, AlphaFold3 (academic), JAX
Docking & chem	AutoDock Vina, DiffDock, Meeko, RDKit, Open Babel, micromamba env
Pocket / MD	fpocket, P2Rank, OpenMM, GROMACS
Variant effect / PGx	AlphaMissense, ESM1b, PharmGKB / CYP450 tables
Drug & ageing data	DrugBank, ChEMBL, ZINC, GenAge/CellAge (HAGR), (optional IMPPAT)
Biological-age ML	PyTorch, scikit-learn, methylation arrays, retinal-image CNN, voice models
Backend	Python, FastAPI, Celery, PostgreSQL + pgvector, Redis
Frontend	Next.js / React, Tailwind, shadcn/ui
Compute	Local Apple-Silicon (dev) → Modal / Lambda Labs / RunPod GPU (scale)
Data governance	DPDP-Act-compliant consent management, audit logging, encryption at rest
Infra/DevOps	Docker, GitHub Actions, Vercel (web), object storage for structures

6. Proof-of-concept — already built (the credibility slide)

Run on a MacBook (Apple M4), zero cloud cost:

A. Folded the rapamycin-binding (FRB) domain of human mTOR

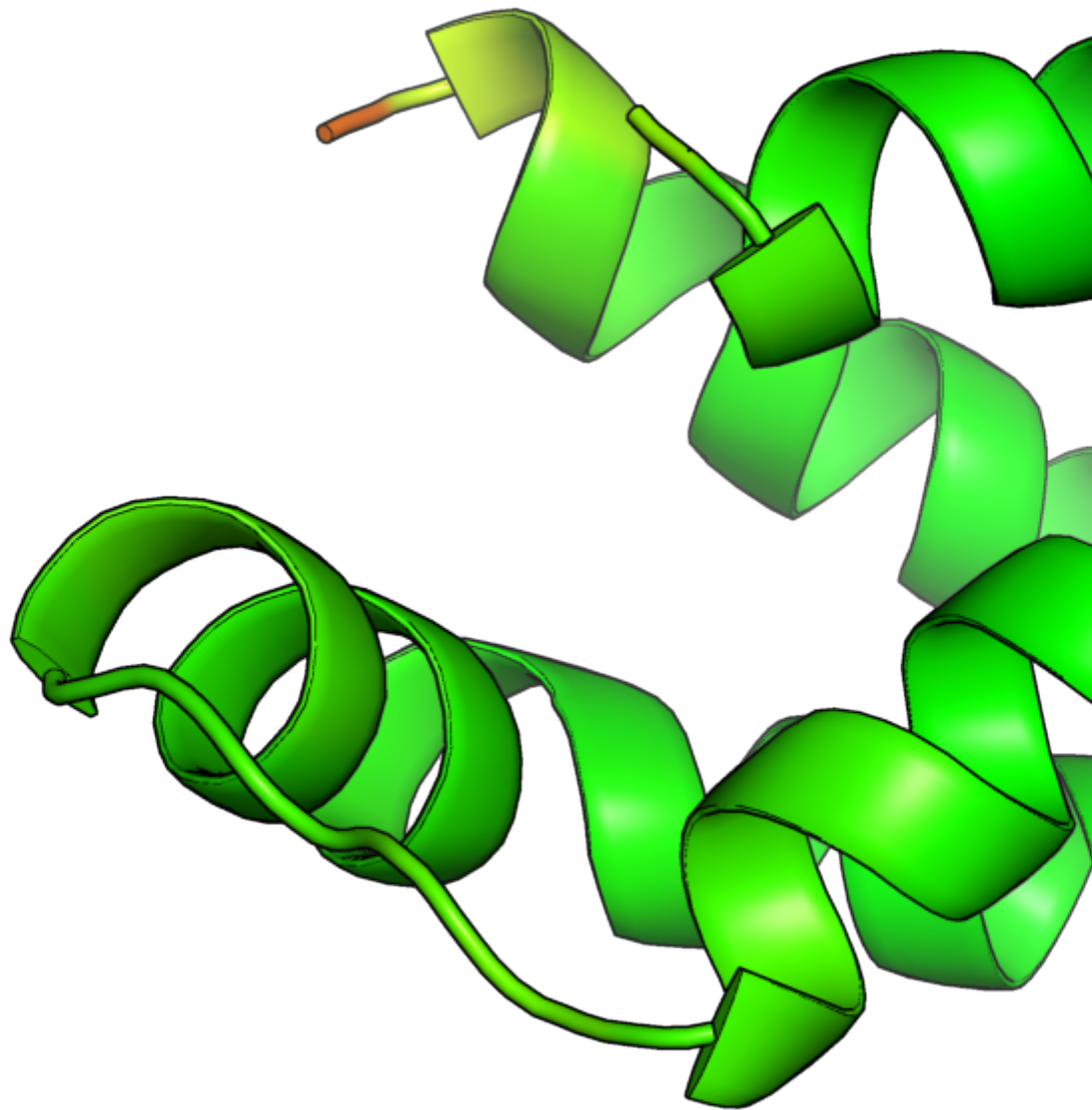
- AlphaFold2, 5 models, mean **pLDDT 92.8** (highest-confidence tier).
- Ser2035 — the residue that anchors rapamycin — folded at **pLDDT 94**.

B. Docked two approved drugs into their targets — with redocking validation

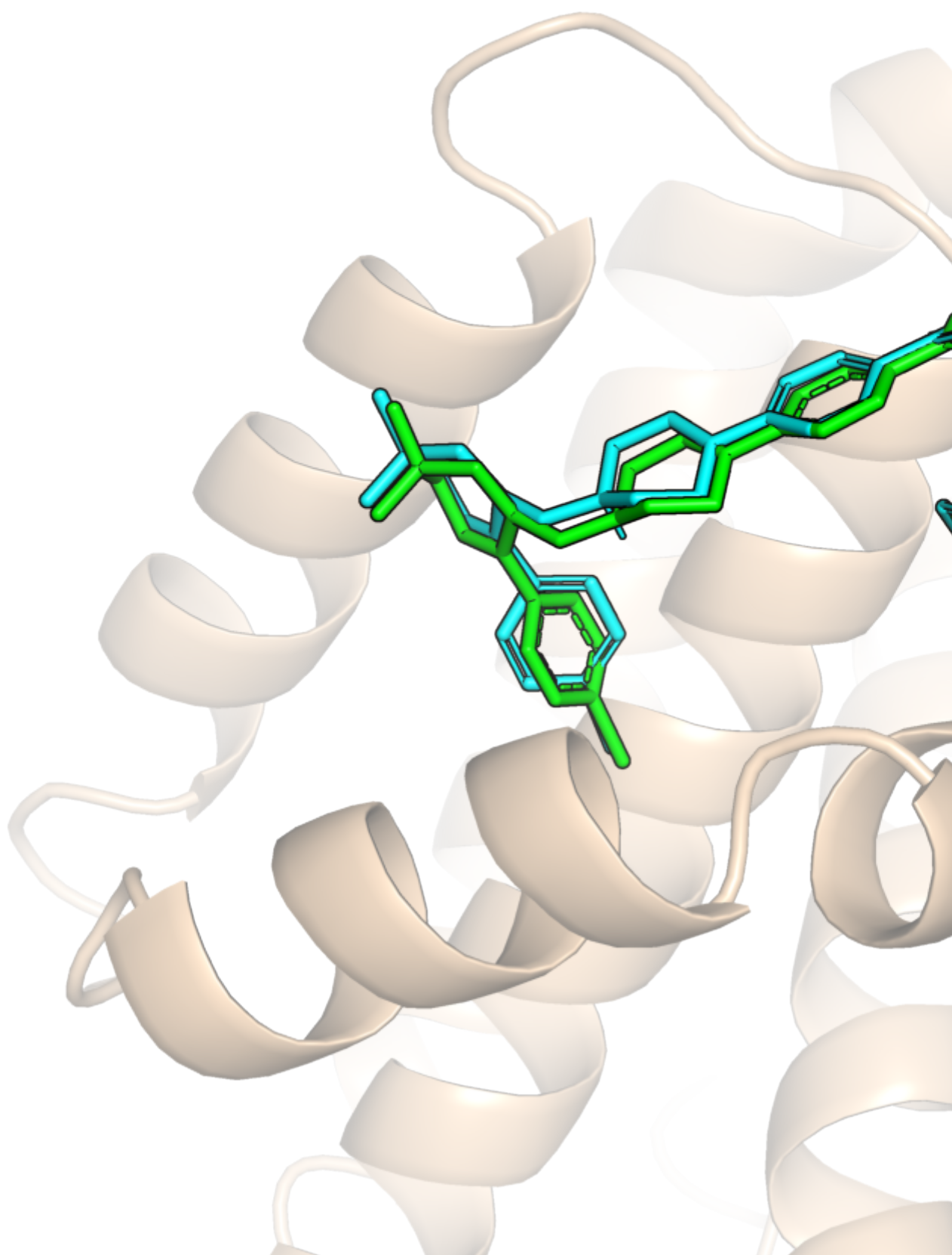
Drug (status)	Target	Binding affinity	Redock RMSD vs crystal
Rapamycin (approved)	FKBP12	-11.09 kcal/mol	0.53 Å
Navitoclax (clinical senolytic)	Bcl-xL	-10.56 kcal/mol	0.90 Å

Both redock RMSDs are **well under the 2 Å success threshold** → the pipeline reproduces experimental drug binding. The affinities are strongly negative → consistent with real potent drugs.
The engine works and its numbers are trustworthy.

Figures (rendered locally):







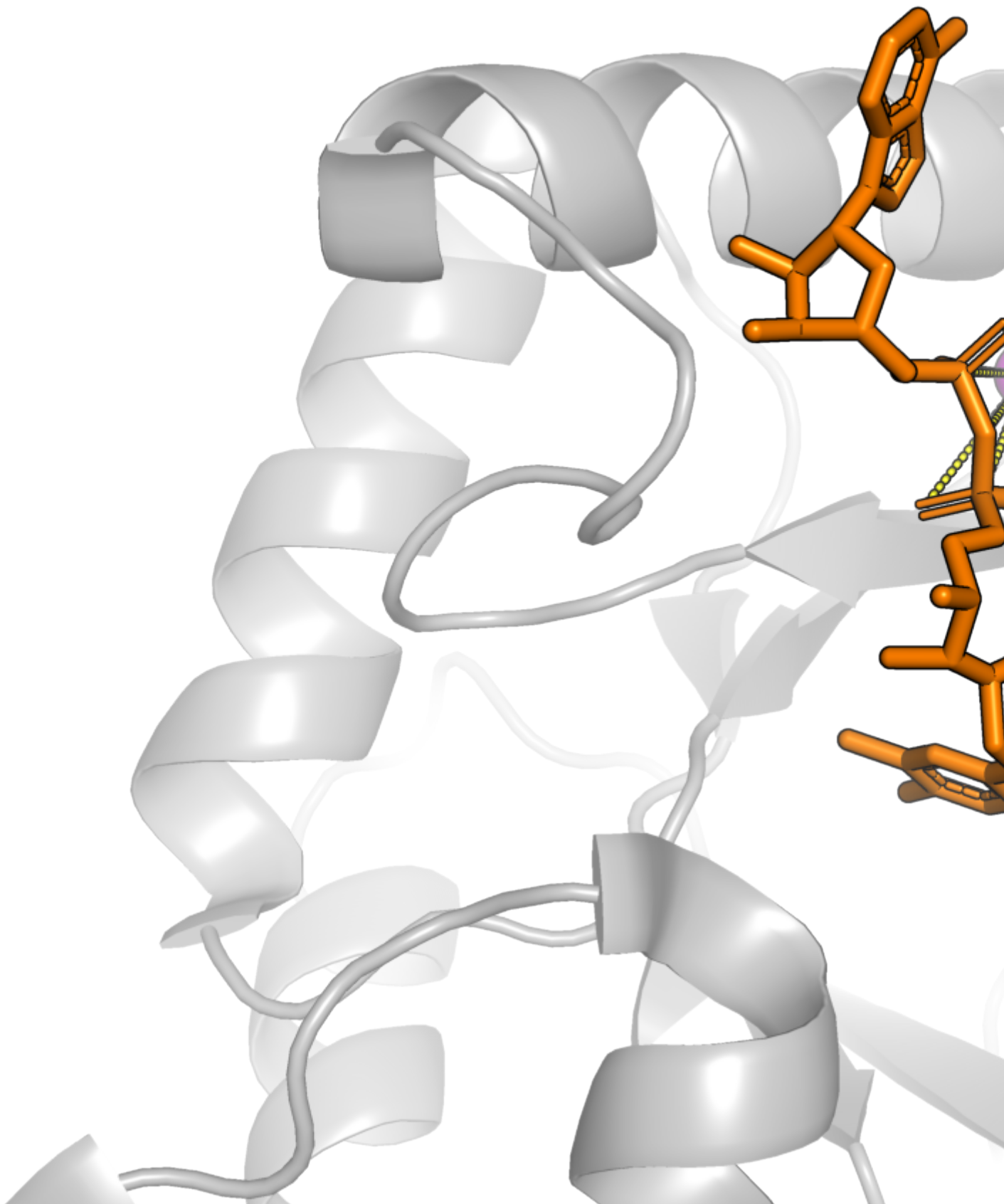
6b. The South-Asian wedge — a *variant-triage* result (MTHFR C677T)

We ran the **MTHFR C677T variant** (Ala222Val), carried at high frequency in South-Asian populations and a driver of hyperhomocysteinemia, cardiovascular ageing, and impaired DNA methylation (the same chemistry epigenetic clocks measure).

Experiment: fold/prepare wild-type vs A222V MTHFR (identical prep, one residue changed), then redock the FAD cofactor into both.

Receptor	FAD binding affinity
Wild-type MTHFR	−11.11 kcal/mol
A222V (C677T) variant	−11.07 kcal/mol

Result: no change in FAD binding affinity — and that is the *scientifically correct* answer. Structural analysis shows residue 222 is **10.1 Å from the FAD pocket** and **buried in a packed hydrophobic core** (30 atomic contacts). C677T is therefore a **protein-stability variant** (bulkier Val destabilises the fold → FAD is lost over time), **not a binding-site variant**. Its matched intervention is **riboflavin/methylfolate** — cheap, approved.



Why this matters more than a clean differential: a naive "dock-drugs-into-every-variant" pipeline would report this null and call C677T harmless — *wrong*. AMRITA's real moat is **variant triage**: classifying each variant as *pocket* / *buried-stability* / *allosteric* and routing it to the correct method (docking vs $\Delta\Delta G$ -stability vs molecular dynamics). This methodological correctness is what survives technical due diligence — and the **$\Delta\Delta G$ -stability + MD module** is the concrete next build.

7. Business model

Line	Customer	Revenue
B2C	Individuals	Biological-age testing + subscription re-measurement
B2B2C	Life/health insurers	Biological-age-based underwriting & wellness
B2B	Longevity clinics, supplement brands	API access to the measurement + matching engine
Pharma/ biotech	Drug developers	License the South-Asian ageing cohort + repurposing hits as trial endpoints

8. Regulatory & go-to-market path

1. **Wellness/diagnostic entry** — biological-age measurement is a wellness/diagnostic product (fast path, CDSCO nutraceutical/diagnostic rules).
2. **Off-label, physician-supervised** repurposing of already-approved drugs (established safety) — generates outcome data legally.
3. **New-indication / dosing-regimen patents** on old drugs (method-of-use IP).
4. Founder edge: **lawyer-technologist** navigates consent (DPDP Act), off-label frameworks, and biobank governance — a rare, defensible skill set for exactly this company.

9. Moat

1. **Proprietary South-Asian ageing-outcomes dataset** (compounds per patient — uncopyable).
2. **India-calibrated clock + variant-aware engine** (model + data IP).
3. **Regulatory/legal navigation** as founder superpower.
4. **Full-stack loop** — measure → match → prove — that no single-point competitor owns.

10. Funding ask & use of funds (illustrative pre-seed)

Raise: pre-seed (sized to milestones below). **Use of funds:**

- Scale the engine to 25–50 ageing targets × full approved-drug library.
- Build BharatAge MVP (retinal + DBS pilot, 500–1,000 participants).
- Generate the first South-Asian variant result (the differentiator).
- Regulatory + consent architecture.

11. Roadmap / milestones

Phase	Milestone	Proves
0 (done)	Fold + dock + validate PoC	Engine works
1	Batch-screen full target × drug matrix	Discovery at scale
2	First South-Asian variant differential result	The wedge is real
3	BharatAge measurement MVP + pilot cohort	The data moat starts
4	First measure→match→prove outcome readout	The flywheel turns

12. Risks & honest mitigations

Risk	Mitigation
Docking ≠ clinical efficacy	Engine is a <i>prioritiser</i> ; truth comes from the measure→prove loop
Large-ligand docking is hard	Validated by redocking RMSD; flag low-confidence hits
Biobank consent/ethics	Founder is a lawyer; DPDP-compliant consent from day 1
Solo, non-PhD founder	Recruit a scientific advisor; PoC proves execution capability
Regulatory uncertainty	Lead with wellness/diagnostic + already-approved drugs

Document generated alongside a working proof-of-concept (mTOR fold + two validated docks) on local hardware.

AMRITA — Pre-Seed Pitch Deck

Slide 1 — Title

- **AMRITA**
- *Longevity medicine, calibrated for India.*
- Computational drug-repurposing for human ageing, built on South-Asian biology.
- Working names under review: Ayuh Bio, Chiranya, GeroBharat.

Slide 2 — Problem

- Ageing is now a targetable biological process — but every major longevity drug was validated on Western populations.
- South Asians age on a different curve: the "thin-fat" phenotype drives insulin resistance, diabetes and heart disease 5–10 years earlier.
- Western drug-response data does not transfer; dosing, efficacy and side-effect profiles miss Indian biology.
- Result: a 1.4-billion-person market has no longevity-drug discovery engine built for it.

Slide 3 — Why Now

- AlphaFold (2021) and AlphaFold3 (2024) made high-quality protein structures free.
- Docking, AI variant-effect prediction and MD validation are now laptop-scale and almost free.
- TAME (metformin), rapamycin and senolytic trials made drug-repurposing for ageing scientifically respectable.
- The shift from blockbuster R&D to repurposing + real-world evidence fits India's cost and data advantages.

Slide 4 — Why India

- **Different ageing biology** — earlier cardiometabolic disease at lower BMI.
- **World's pharmacy** — largest generic-drug manufacturing base; any repurposed geroprotector can be produced and distributed cheaply.
- **Massive, under-studied population** — no omic-grade ageing biobank exists for 1.4 billion people.
- **Cost arbitrage** — dry-lab discovery + cheap diagnostics + low-cost trials.

Slide 5 — The Product: Measure → Match → Prove

- **BharatAge** — biological-age clock calibrated to Indian physiology, from free face/voice tiers to dried-blood-spot methylation + metabolomics.

- **The Engine** — AlphaFold + docking screens the approved-drug library (DrugBank/ChEMBL) against ageing targets, re-ranked by personal genotype.
- **Outcomes loop** — physician-supervised matched drugs, biological age re-measured in 3–6 months.
- Output: a proprietary, compounding dataset of *which approved drug moved biological age in which South-Asian phenotype*.

Slide 6 — The Engine + Proof-of-Concept

- Built a fully local dry-lab pipeline: structure prediction → pocket detection → docking → redock validation → variant personalization.
- **mTOR FRB domain folded at mean pLDDT 92.8**; rapamycin-anchoring Ser2035 at pLDDT 94.
- Rapamycin → FKBP12: **−11.09 kcal/mol**, redock RMSD **0.53 Å**.
- Navitoclax → Bcl-xL: **−10.56 kcal/mol**, redock RMSD **0.90 Å**.
- Both RMSDs are well under the 2 Å success threshold — the engine reproduces known binding and its numbers are trustworthy.

Slide 7 — The South-Asian Wedge: variant triage

- **Demonstrated on MTHFR C677T (Ala222Val)** — high-frequency South-Asian variant driving homocysteine, cardiovascular ageing, and impaired DNA methylation (the chemistry epigenetic clocks measure).
- Our engine folded WT vs variant and redocked the FAD cofactor: **no change in binding affinity (−11.1 vs −11.1)** — and that is the *correct* answer. Residue 222 is 10 Å from the pocket, buried in the core: **C677T is a stability variant, not a binding variant**. Fix = riboflavin (cheap, approved).
- **The moat is variant triage** — classifying each variant (pocket / stability / allosteric) and routing it to the right method (docking vs $\Delta\Delta G$ -stability vs MD). Naive pipelines report a false "harmless" and fail technical diligence.
- First mover on a variant-aware, India-calibrated geroprotector engine — built to *not overclaim*, which is what survives investor due diligence.

Slide 8 — Business Model

- **B2C**: Biological-age testing + subscription re-measurement.
- **B2B2C**: Life/health insurers — age-adjusted underwriting and wellness incentives.
- **B2B**: Longevity clinics, supplement brands — API access to measurement + matching engine.
- **Pharma/biotech**: License repurposing hits + South-Asian ageing cohort and real-world endpoints.

Slide 9 — Moat

- **Proprietary South-Asian ageing-outcomes dataset** — compounds × patient × biological-age change; uncopiable once scaled.
- **India-calibrated clock + variant-aware engine** — model + data IP.

- **Regulatory and consent architecture as founder superpower** — DPDP-compliant biobank governance from day one.
- **Full-stack loop** — measure → match → prove — no single-point competitor owns the entire flywheel.

Slide 10 — Regulatory Path

- Enter through **wellness/diagnostic biological-age testing** — fast path under existing CDSCO diagnostic/nutraceutical rules.
- Enable **off-label, physician-supervised repurposing of already-approved drugs** to generate outcome data legally.
- File **method-of-use / new-indication patents** on old drugs for ageing-related dosing regimens.
- Lawyer-technologist founder de-risks consent, off-label frameworks, biobank governance and DPDP Act compliance.

Slide 11 — Team

- **Solo founder — lawyer-technologist.**
- Built the working proof-of-concept end-to-end on an Apple Silicon laptop: mTOR fold + two validated docks, zero cloud spend.
- Legal background is a rare, defensible edge for a company that must navigate patient consent, off-label use, biobank governance and drug-regulatory frameworks.
- Now raising pre-seed to add scientific advisors, computational biologists and a clinical operations lead.

Slide 12 — The Ask

- **Raising a pre-seed round** to reach first South-Asian variant result and BharatAge pilot.
- **Use of funds:**
 - Scale the engine to 25–50 ageing targets × full approved-drug library.
 - Build BharatAge MVP (retinal + dried-blood-spot pilot, 500–1,000 participants).
 - Generate first South-Asian variant differential result.
 - Build regulatory, consent and data-governance architecture.
- **Milestones:**
 - 0–3 months: batch-screened target × drug matrix.
 - 3–6 months: first variant-aware hit prioritisation.
 - 6–12 months: BharatAge pilot launched + first measure→match→prove readout.

Appendix — The Build: Prompt Library

A sequenced set of prompts to build AMRITA toward a fundable demo. Each is written to be pasted into an AI coding agent (Claude Code). Run them roughly in order; each phase produces a tangible artifact for the pitch.

Phase 0 — Science engine (✅ mostly done this session)

P0.1 — Fold an ageing target

"Using LocalColabFold at ~/localcolabfold, fold the [DOMAIN] of [PROTEIN] (UniProt [ID]). Extract the domain sequence from UniProt, run 5 models, and report mean pLDDT and the confidence at the key catalytic/binding residue."

P0.2 — Validate a docking pipeline by redocking

"In the `dock` micromamba env, redock the native ligand [CODE] from PDB [ID] into its receptor: extract receptor + native ligand, prep with Open Babel/Meeko, build the search box from the native ligand centroid, run AutoDock Vina (exhaustiveness 16), then report binding affinity and the redock RMSD vs the crystal pose using obrms."

P0.3 — Screen approved drugs against one target

"Download the approved-drug subset of DrugBank (SMILES). For target [PROTEIN] structure, prep each ligand with Meeko, dock all with Vina against the validated pocket, and output a ranked table of binding affinities. Flag the top 20 and any known geroprotectors."

Phase 1 — Engine at scale (the discovery product)

P1.1 — Target database

"Build a PostgreSQL schema for ageing targets: protein, UniProt ID, structure path, pocket coordinates, pathway, GenAge/CellAge annotation. Seed it with mTOR, Bcl-2, Bcl-xL, SIRT1-7, AMPK, NLRP3, KEAP1, FOXO3, Klotho."

P1.2 — Automated fold→pocket→dock pipeline

"Write a Python pipeline (Celery + FastAPI) that, given a target and a drug library, runs: ESMFold (or cached AlphaFold) → fpocket pocket detection → Vina batch docking → results to Postgres. Make it resumable and parallel."

P1.3 — Hit ranking + report

"Build a ranking module that scores (target × drug) pairs by affinity, druggability, known geroprotector status, and clinical-approval level. Generate an HTML report per target with top hits and binding-pose thumbnails."

P1.4 — Validation harness

"For every new target, automatically redock a known co-crystallised ligand and store the RMSD as a per-target confidence score. Block hits from targets that fail validation (RMSD > 2.5 Å)."

Phase 2 — The South-Asian differentiator (the wedge)

P2.1 — Variant-aware folding

"Given a list of coding variants common in South-Asian populations (from GenomeAsia/IndiGen) in an ageing target, generate the mutant sequence and fold it with ESMFold. Compare the binding pocket to the wild type (RMSD + pocket-volume change)."

P2.2 — Differential drug binding

"Re-dock the top drug hits against both wild-type and South-Asian-variant structures of the target. Output drugs whose predicted binding changes most by genotype — these are the personalised/precision candidates."

P2.3 — Pharmacogenomic layer

"Integrate CYP450 metaboliser status (from PharmGKB) for South-Asian allele frequencies. For each candidate drug, annotate predicted metabolism (poor/normal/rapid) to flag dosing concerns."

Phase 3 — BharatAge measurement MVP (the moat starts)

P3.1 — Retinal-age model

"Build a CNN (PyTorch) that predicts age from retinal fundus images using an open dataset; expose the 'retinal age gap' (predicted minus chronological) as a biological-age proxy. Wrap it in a FastAPI endpoint."

P3.2 — Multimodal clock

"Combine face-age, voice-age, grip, and questionnaire features into a single biological-age estimate with a calibration layer that can be re-fit on Indian data. Output uncertainty bounds."

P3.3 — Consumer app

"Build a Next.js app: user uploads a selfie + retinal photo + answers a short form, gets a biological-age dashboard vs the Indian curve, and a personalised report. DPDP-compliant consent flow and encrypted storage."

P3.4 — Outcomes loop

"Add longitudinal tracking: store baseline biological age, the matched intervention, and re-measurement at 3/6 months; compute the delta and aggregate anonymised cohort-level results."

Phase 4 — Fundraising materials

P4.1 — Pitch deck

"From AMRITA_company.md, generate a 12-slide pre-seed deck: problem, why-now, why-India, the engine (with the PoC validation numbers), the measure→match→prove loop, moat, business model, regulatory path, team, ask, use of funds, milestones."

P4.2 — Demo money-slides

"Render publication-quality figures: (1) the folded mTOR FRB domain coloured by pLDDT, (2) rapamycin redocked into FKBP12 with the 0.53 Å overlay, (3) navitoclax in Bcl-xL at 0.90 Å. Use PyMOL (pymol-open-source)."

P4.3 — One-pager + data room

"Produce a one-page executive summary and a data-room checklist (cap table, IP/patent strategy memo, regulatory memo, consent framework, technical validation report)."

P4.4 — Investor list + outreach

"Draft a target list of India deep-tech + global longevity VCs, and a warm-intro outreach email tailored to a lawyer-technologist solo founder with a working dry-lab engine."

Suggested order

P0 (done) → P1.1–1.4 → P2.1–2.2 → P4.1–4.2 (deck with the wedge) → raise → P2.3, P3.* with funding.

Principle: build only as far as the *next* fundraising conversation requires. The engine + one South-Asian variant result + a clean deck is enough to raise pre-seed. Don't build the full app before the cheque.